

To: EML 4501C Senior Design

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Technology Study Memo for Data Handling & Sensor Integration

Summary of Memo

The main task of the Hybrid aerial water drone is as follows: fly out over a body of water, submerge and complete any task desired, then resurface to the initial launch location.

This memo examines three primary areas: communication systems, sensor selection, and system constraints related to power consumption and available space. The goal is to identify technologies that will support reliable operation while remaining compatible with the project's weight, power, and packaging limitations.

Topics for Research

- How will communication be maintained during operation?
 - Communication limitations in water.
- Comparison of communication methods:
 - Wi-Fi
 - Radio telemetry
 - Acoustic communication
 - Tethered communication
- Data rate and bandwidth requirements for:
 - Sensor telemetry
 - Video transmission

- Packaging constraints and available power
 - What sensors are required for navigation, stabilization, etc?
 - Comparison of essential and optional sensors
 - Power consumption of individual sensors and communication systems.

Communication Systems and Data Transmission

Communication is vital to the drones' operations, but is challenging because systems that work in air do not work well underwater. In the air, the drone can use standard radio communication methods such as Wi-Fi or telemetry to send sensor data, video, and control signals back to the operator [2]. These systems can support relatively high data rates, which makes real-time monitoring possible during flight. However, once the drone enters the water, these same signals weaken quickly and may stop working completely. Because of this, underwater communication may need to rely on slower methods like acoustic signals or may not be practical at all for continuous use.

Due to these limitations, the drone will likely need to store some data onboard while underwater instead of sending it live. Once the drone resurfaces, it can transmit stored data such as sensor readings and video back to the operator. This means the system must include reliable onboard storage and be able to continue operating safely even when communication is temporarily lost. Additionally, critical sensors such as cameras or altimeters may still have to provide live communications in order to operate the drone effectively.

Wi-Fi and radio telemetry both work well in air, but they have different strengths. Wi-Fi can send large amounts of data, such as video, but it usually has a shorter range and is

more sensitive to interference. Radio telemetry has a longer range and is more reliable for basic control signals, but it cannot transmit large amounts of data like high-quality video. Because of this, many drone systems use both: telemetry for control and Wi-Fi or similar systems for data when available. However, once the drone enters the water, these same systems become unreliable. Water blocks radio frequency signals very quickly, meaning both Wi-Fi and telemetry stop working almost immediately after submersion.

Other methods can be considered, but each has limitations. Acoustic communication can work underwater because it uses sound instead of radio waves, but it is much slower and cannot support high data rates like video. It is mainly useful for sending simple messages or basic sensor updates. A tethered connection could provide reliable communication in both air and water, but it limits how far the drone can travel and adds mechanical complexity, which may not fit the mission requirements.

Sensor Selection and Integration

The drone requires sensors to understand its movement, position, and environment. Some sensors are necessary for basic operation, while others are optional and depend on mission needs. At a minimum, the system will need sensors that measure motion, depth, and overall system health. These include motion sensors, a pressure sensor for underwater depth, and a camera for visual navigation or mission tasks. Battery monitoring is also important so the system can track available power.

Additional sensors such as GPS, sonar, or range sensors could improve performance, especially for navigation and obstacle detection.

Table 1 | Levels of autonomy: requirements, availability and readiness for market

	Exteroceptive sensors	Computational load	Supervision required	Readiness level	Validated on drone type
Sensory-motor autonomy	None or few	Little	Yes	Deployed	All types
Reactive autonomy	Few and sparse	Medium	Little	Partly deployed	Fixed wing, rotorcraft and flapping wing
Cognitive autonomy	Several and high density	High	None	Not yet deployed	Mostly rotorcraft

Figure 1: Higher levels of autonomy require more sensors and greater processing capability. Adapted from Floreano and Wood [3].

However, each added sensor increases complexity, power use, and space requirements. Because of this, the system should prioritize a small, reliable set of core sensors first, and only add extra sensors if they clearly improve mission performance.

Data Rate and Bandwidth Requirements

Different types of data on the drone require very different amounts of communication bandwidth. It is important to decide what can be transmitted live and what must be stored onboard, especially during underwater operation. Sensor telemetry refers to basic data from onboard sensors such as position, depth, and system status. This type of data is relatively small and does not require high bandwidth. In most drone systems, telemetry is typically sent in short, regular updates rather than continuous high-speed

streaming. Because of this, telemetry is the most realistic type of data to transmit in real time when communication is available.

Video transmission, however, requires significantly more bandwidth. A camera stream must send large amounts of image data every second. This makes video the most demanding part of the communication system. While video can be transmitted in air using Wi-Fi or similar systems, it becomes impractical underwater due to the lack of reliable communication methods. Available options include a tether connected to the drone, or a surface buoy containing a wifi transmitter that broadcasts the signal for the camera feed [1]. Because of this difference, the system design will likely need to treat these two data types separately.

Packaging Constraints and Available Power

The drone design is heavily limited by both available internal space and available power, which directly affects what sensors and communication systems can realistically be included. Since the vehicle must contain batteries, propulsion systems, waterproof housing, and control electronics, the remaining internal volume for sensors is expected to be limited. This means sensor selection must prioritize compact size and efficient integration, rather than trying to include every possible sensing option. Placement is also important, since some sensors may need external exposure, while others must remain fully sealed inside the vehicle.

From a functional standpoint, a minimum set of sensors is required for operation in any environment. These essential sensors include an IMU to measure motion and orientation, a magnetometer for heading direction, a pressure sensor to determine

underwater depth, a camera for visual feedback, and a battery monitoring system to track available power. These sensors form the baseline system that ensures the drone can operate safely in both environments.

Power consumption is another major constraint that directly influences sensor selection. All sensors and communication systems draw power from the same battery that also supplies the propulsion system, which is expected to be the largest energy consumer. As a result, only a limited portion of total energy is available for sensing and data handling systems. Low-power sensors such as the IMU and pressure sensor can operate continuously with minimal impact on overall mission duration.

However, higher-power components such as cameras and communication systems can significantly reduce operating time if used continuously. Essential sensors should remain active throughout the mission, while higher-power systems should be used only when necessary or operated in reduced-power modes when possible.

Conclusion

Overall, the system is limited by communication, space, and power constraints due to the need to operate in both air and underwater environments. Communication is reliable in air using Wi-Fi or telemetry but becomes largely unavailable underwater, requiring onboard data storage and delayed transmission once the drone resurfaces, a tether, or surface buoy. The system therefore requires a small, essential sensor suite for stable operation, including motion sensing, depth measurement, direction tracking, visual input, and battery monitoring, while optional sensors should only be added if they clearly improve mission performance. At the same time, internal space is limited by core

components such as batteries and propulsion systems, and power is heavily consumed by propulsion, leaving only a small budget for sensors and communication. Because of these constraints, all sensing and data systems must be compact, efficient, and carefully prioritized. Successful design depends on keeping the system simple and reliable while balancing performance with strict physical and power limitations.

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Summary

The goal of our senior design project is to design and build a hybrid aerial-underwater drone that can be piloted to the target location efficiently, seamlessly transition between the air and ocean surface and dive down to the research area. It will then use sonar and other sensors to image and retrieve data, and then transition back to the air and fly back to the launch platform to transfer the results. A few of the problems that this technology can be used to solve include coral reef research, tagging of sea creatures, and inspections of oceanic machinery or vessels.

The purpose of this study memo is to familiarize us with past work and information relevant to the propulsion and fluid dynamics system of the project. The research that has been conducted will be organized and consolidated to help us easily identify existing technologies and provide details on the parameters, capabilities, and uses of the technologies that are already available in this system.

Technologies and questions to address

- Underwater vs aerial propulsion design
- How does the length of a propeller blade affect the performance?
- How does the speed of rotation affect performance?
- Are propeller blades more prone to damage while underwater?
- Propeller configurations
- Tilting propeller designs (Bell Boeing V-22 Osprey, Leonardo AW609)
- Simulation and testing techniques

Methodology

Research Techniques: Research was conducted on the web by using specific search terms, such as “propeller fluid dynamics”, “propeller configurations”, and “hybrid propeller capabilities”, to find irrefutable sources in the form of technical reports, aerospace engineering journals, mission documents, and more. By using these search terms and sources, I was able to find a wide variety of technologies that have been used in the same or comparable way to the functions that are needed in this project. I ensured to pick technologies that have been backed by adequate research and testing, rather than technologies that do not exist yet or are hypothetical. Though theoretical papers are intriguing and often backed with irrefutable evidence, this memo will focus only on existing technologies.

Parameters Compared:

- Air vs Water Performance - Fluid density, viscosity, thrust and torque requirements, Reynolds number, RPM requirements
- Propeller configurations - pitch angle, blade length, 3 vs 4 rotor design
- Propeller designs - fixed vs variable pitch, folding vs ducted, normal vs contra-rotating propellers (may not dive deep into in this memo)
- Tilting propulsion - fixed vs tilting propeller efficiency and performance

- Simulation (CFD) Techniques - ANSYS Fluent / CFX, STAR-CCM+, OpenFOAM

Background and Advantages: As described before, hybrid water-aerial drones are drones that are able to fly in both water and the air, transitioning between the two in whichever way a design team chooses. For our project, we are likely going to implement an autonomous design for the transition, but that will be detailed in another tech memo. While there are drawbacks to using a hybrid drone over a drone specifically designed for air or water travel, there are also many positives. While a drone designed for only water may be able to perform better and more efficiently than its hybrid counterpart, it lacks the ability to avoid unwanted areas of the water (strong currents or hazardous zones), as well as the ability to avoid dangers (predatory creatures) by flying above the water when the threat is detected. Another advantage of hybrid drones is that it eliminates the need for a group of people to put themselves in a dangerous situation. Areas such as coral reefs and shark infested waters can be very dangerous if the researcher is not highly skilled and experienced. This leads to the advantage that this drone removes the need to be SCUBA certified if the area you wish to study requires the certification.

Results

Underwater vs. Aerial Propulsion Parameters

	Air (Sea Level)	Water (Sea Level)
Density	1.229 kg/m ³	1025 kg/m ³

Specific Volume	0.814 m ³ /kg	9.76x10 ⁻⁴ m ³ /kg
Pressure	101.3 kN/m ²	101.3 kN/m ²
Temperature	15 degC, 288 K	15 degC, 288 K
Viscosity	1.73x10 ⁻⁵ N-s/m ²	1.19x10 ⁻³ N-s/m ²
Kinematic Viscosity	1.46x10 ⁻⁵ m ² /s	1.14x10 ⁻⁶ m ² /s
Gas Constant	0.286 J/g-K	incompressible
Specific Heat	0.715 J/g-K	3.99 kJ/kg-K
Ratio	1.4	none
Reynolds Number (at same length and velocity)	~68,493	~877,192
Required Torque	Low - Medium	High
Required RPM	Very High	Low

Propeller Configurations

Pitch angle: Pitch angle is the angle between the blade chord line and the plane of rotation and determines how much air or water the propeller can move during each 360 degree revolution. As a general rule of thumb, a higher pitch angle results in more movement per revolution than a lower pitch angle, but also requires more power to operate [1]. When considering the effects of pitch angle in water vs air, it is apparent that the optimal angle for efficiency varies largely. The best metric to use for optimizing pitch angle is the pitch to diameter ratio (P/D) since the angle of pitch changes along the blade. The optimal pitch ranges for air and water are as followed:

Environment	Pitch Angle	P/D ratio
Air	10-20 deg	0.3-0.7
Water	25-35 deg	0.9-1.2
Both	20-25 deg	0.7-0.9

Although meeting in the middle is theoretically a good way to optimize the transition and allow for both air and water travel, this can lead to issues further down the road with power efficiency, stalling, cavitation, rotor loading, and many other factors. A potential solution to this problem will be expanded upon in the “Fixed vs Variable Pitch” section.

Blade Length: Propeller blade length/diameter has a massive impact on the thrust that a propeller system can generate. This is shown by the thrust equation $T \propto D^4$. In theory, increasing the diameter of the propellers by a small amount can have a huge impact on the thrust generated. By using the swept area formula for a single propeller system, $A = (\pi D^2)/4$, we see that increasing the diameter also increases the area, which in return increases the amount of fluid that is being accelerated beneath [1]. Though these positives sound great, the drawback is that this increase in diameter also increases the power and torque required to run the system. Finding the optimal diameter of our propellers will vastly depend on the power that is available. The power budget for our drone will be determined later on in the project process.

3 vs 4 Rotor Design: Quadcopters (4 rotors) tend to be the most widely used layout due to their simple controls and very high stability. Though they are the industry standard, tricopter (3 rotors) designs have been widely used as well. The optimal number of propeller/rotor systems for this drone will be determined once we decide on if we will be implementing the tilting propulsion technology described later in the memo. The pros and cons of 3 vs 4 rotor designs are tabled below.

	Pros	Cons
4 Rotors	High stability and control Widely available parts More affordable parts Simple flight controls	Higher battery consumption More resources Extra part to maintain Not much room for innovation
3 Rotors	Higher efficiency Tail propeller allows for yaw control Allows for more innovation	More complex due to tail servo Odd number of rotors Higher vibrational load

Propeller Designs

Fixed vs Variable Pitch: After looking into the importance of pitch angle we are presented with an opportunity to implement a certain technology known as variable pitch propeller blades. This technology would allow us to change the pitch angle during the transition from the air to water and back. Implementing a variable pitch design allows us to maximize the pitch angle efficiency for both the air and water environments without many

drawbacks. While implementing this technology would mean adding an extra system to the drone that may require more power, it would also maximize flight efficiency and save more power than what is lost [2].

Tilting Propulsion

Fixed vs Tilting propulsion: A fixed propulsion system is a design choice in which the rotors and propellers remain mounted in an upright position. Due to the fixed nature, all maneuvers must be done by changing the rotating speed of individual propellers. This usually implies that the speed of the propellers in the direction you direct the drone to go is slower than the opposite side. For example, if you want the drone to move to the right, the rotors on the left side will speed up or the right side will slow down. Tilting propulsion systems operate by the assemblies rotating about a pivot point [3]. This allows for the thrust to be aimed however seen fit, which can greatly increase the speed and efficiency of the system [3]. An example of this is that the propellers may remain horizontal during launch and the flight to the specified location, then turn to a vertical orientation once underwater. This may greatly increase the efficiency and maneuverability of the drone. One final note is that tilting propulsion can also be used in 4 rotor designs, not just the 3 rotor designs detailed in the next section.

Propulsion Type	Advantages	Disadvantages
Fixed	Simple design Low weight Easier waterproofing Less chance of failure (reliable) Less expensive	Less efficient Harder to optimize transition zone
Tilting	Optimize for each environment Increased maneuverability Smoother transition between environments Increased efficiency in air Greatly increased efficiency in water Looks cool (had to add that)	Mechanically complex More expensive More systems to maintain Heavier Difficult to waterproof Failure more likely

Pre-existing Tilting Configurations with Unique Designs: Two examples of vehicles that use a mix of the tilting and <4 rotor technologies on a larger scale include the Bell Boeing V-22 Osprey and the Leonardo AW609. These interesting vehicles implement a unique configuration being two tilting propellers on the sides and an airfoil on the back to stabilize the aircraft. These have the advantages listed above but have one less propeller to maintain, while also having benefits of a wing-supported cruise [4]. This

increases the endurance of the drone by greatly increasing range and time that they can spend in the air [5].



Image 1: Bell Boeing V-22 Osprey
Source: Adapted from [4]



Image 2: Leonardo AW609
Source: Adapted from [5]

Although this specific configuration has not, to my knowledge, been used in a hybrid drone design, these illustrate that unconventional designs can lead to successful outcomes. A similar configuration to these, perhaps even using a 3rd propeller rather than an airfoil, could prove beneficial in the design of our drone.

Simulation Softwares & Techniques

Software	Pros	Cons
ANSYS Fluent / CFX	Industry Standard Free for students Easy to import geometry Great for transition modeling	Difficult to learn Requires above average hardware Limited API Free learning resources
STAR-CCM+	Best moving-body sims. Strong surface modeling Used widely in amphibious research Free for students in clubs	Difficult to learn UI considered overwhelming Fewer free learning resources
OpenFOAM	Free for anyone Code your own simulations Open source Flexible	Low precision if not done correctly Requires a strong coding background

Conclusion & key Takeaways

- Propellers in water require a higher torque and lower RPM than in air.
- The optimal fixed blade pitch angle that will be sufficient in both environments is between 20-25 degrees with a P/D ratio from 0.7-0.9.

- An increase in diameter of propeller blades also increases the power and torque required to run the system. To optimize the drone we must find the best balance.
- 4 rotor designs are more stable and simple than 3 rotor designs but they allow for less innovation and no yaw control.
- Implementing variable pitch allows us to maximize the pitch angle efficiency for both the air and water environments, increasing the total efficiency of the drone.
- Tilting propeller systems have many advantages but just as many disadvantages, however, if we can make it work, we may end up with a very efficient and unique drone design.
- Implementing the configuration of the Boeing V-22 Osprey may work, but will need much more research to prove beneficial.

References & Citations

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A drone that intends to operate in both air and water seamlessly requires varying levels of power. In high drag mediums such as water, the necessary torque and power is significantly higher than that of mediums such as air. The weight of the battery is also a critical consideration as a heavy battery might help with the goal of submerging, but would greatly inhibit the drone's

ability to fly. Therefore, the selection of a battery and related electronics is crucial to the drone performing the intended capabilities.

Lithium based batteries offer some of the best performance and cycle life. There are six main lithium battery compositions available. Lithium Iron Phosphate (LiFePO_4), Lithium Cobalt Oxide (LCO), Lithium Manganese Oxide (LMO), Lithium Nickel Manganese Cobalt Oxide (NMC), Lithium Nickel Cobalt Aluminum Oxide (NCA), and Lithium Titanate (LTO) (Sageadmin, 2022). Important metrics when determining the battery that best suits this application include specific energy, energy density, pulse rate capability (C), power density, and specific power. The specific energy refers to the amount of energy contained in a battery relative to its mass. The energy density is a measure of the amount of energy contained in comparison to its volume (Niclas, 2024). The pulse rate capability or C rate describes the rate by which the battery is able to discharge. A higher C rate will mean more power flowing at a given instant from the battery (Abts, 2025). The power density of a battery is the amount of power a battery can provide compared to its size and the specific power describes the amount of power compared to the mass. (Amprius, 2025) These values are pivotal in determining if the battery will be able to provide enough power in both mediums, while maintaining battery life and flight speeds.

The table figure below from Linden's Handbook of Batteries by Kirby W. Beard shows how the chemical composition affects the physical capabilities of the battery. Based on this data, NMC and LMO power cells, with a specific energy of 100 - 175 Wh/kg, a power density of 10,000 W/L, a specific power of 4000 W/kg and a pulse rate capability of over 100 appear to be very capable batteries. These battery compositions also have an energy density second only to energy cells, with NMC batteries having an edge over LMO batteries. These compositions of power cells will allow for a high rate of discharge and as a result, large amounts of power when necessary. When comparing this to the LiCoO_2 and NMC Energy Cells, it is clear that these batteries prioritize longer runtime over high power. The power density for these energy cells are 2000 W/L, the energy density ranging from 370 to 735 Wh/L, the specific energy from 140-275 Wh/kg, a specific power of 1000 W/kg, and a pulse rate capability of 5. When comparing these two battery types, energy cells prioritize energy density with a slower drain rate and lighter weight, meanwhile power cells focus on providing large amounts of power and torque necessary for high load applications. When considering the application for this project, high loads are expected and therefore the NMC or LMO power cells seem like strong contenders.

TABLE 17A.12 General Performance Characteristics of Li-Ion Cells (Cylindrical, Prismatic, and “Polymer”) Using Common Cell Chemistries

Characteristic	LiCoO ₂ /graphite NMC/graphite NCA/graphite Energy cells	NMC/graphite LMO/graphite Power cells	LiFePO ₄ /graphite Power cells	LMO/Li ₄ Ti ₅ /O ₄
Voltage range, V	2.5–4.2 typical 2.5–4.4 for some cells	2.5–4.2	2.5–3.6	2.8–1.5
Avg. voltage	3.7–3.85	3.7	3.3	2.3
Specific energy, Wh/kg	175–275 cylindrical 140–240 polymer	100–175	60–110	70
Energy density, Wh/L	400–735 cylindrical 370–600 polymer	250–460	125–250	120
Continuous rate capability	–3C	Over 30C	10 to 125C	10C
Pulse-rate capability, C	5	Over 100	Up to 250	20
Cycle life at 100% DOD	500+	500+	1000+	4000+
Calendar life, year	>5	>5	>5	>5
Self-discharge rate, %/month	1–5 %/mo	1–5 %/mo	1–5 %/mo	1–5 %/mo
Charge temperature, °C	0 to –45	0 to –45	0 to –45	–20 to –45
Discharge temperature, °C	–20 to +60	–30 to +60	–30 to +60	–30 to +60
Memory effect	None	None	None	None
Power density, W/L (pulse)	–2000	–10,000	–10,000	–2000
Specific power, W/kg (pulse)	–1000	–4000	–4000	–1100

NMC = LiNi_{1/3}Mn_{1/3}Co_{1/3}O₂, LiNi_{0.5}Mn_{0.3}Co_{0.2}O₂ or LiNi_{0.42}Mn_{0.42}Co_{0.16}, etc.NCA = LiNi_{0.8}Co_{0.15}Al_{0.05}O₂, etc.LMO = Li_{1-x}Mn_{2-x}O₄, etc.Figure 1: A comprehensive comparison of the six main types of lithium chemistries used today.
(Beard et al., 2019)

Characteristic	LiCoO ₂ /graphite NMC/graphite NCA/graphite Energy Cells	NMC/Graphite LMO/graphite Power Cells
Specific Energy Wh/kg	175-275 Cylindrical 140-240 Polymer	100-175
Energy Density Wh/L	400-735 Cylindrical 370-600 Polymer	250-460
Pulse Rate Capability C	5	Over 100
Power Density W/L (Pulse)	~2,000	~10,000
Specific Power W/kg (Pulse)	~1,000	~4,000

Table 1: Highlighting key differences between energy cells and power cells.
(Beard et al., 2019)*Battery Chemistry Heat Generation & Management*

The specific heat capacity of various battery types are as follows: “Lithium Nickel Cobalt Aluminium Oxide (**NCA**) = 830 J/kg.K, Lithium Nickel Manganese Cobalt (**NMC**) = 1040 J/kg.K” (Nigel, 2025). This means that of the compositions, the NMC battery would be able to absorb

the most energy before increasing in temperature by 1 degree. There is not much data on the individual heat generation rates for different battery compositions. However, heat dissipation should be a consideration for this project and be accounted for.

Typical Energy Requirements From a Submersible Drone

A submersible drone product lists the standard battery voltage as 14.8 V with 6,500 mAh and a runtime of 2 hours. The same drone claims a 5 hour runtime using a 14.4 V battery with a 14,750 mAh capacity and 25 A max discharge. (*Blueye X3 Underwater Drone: Professional roV for inspection n.d.*)

Electronic Speed Controller Selection

There are numerous electronic speed controllers (ESC) available to purchase with different specifications. The team can opt for a 4 in 1 ESC that controls all four motors concurrently, or for four individual bi-directional electronic speed controllers which would allow for better opportunities for cooling. Bi-directional ESCs will be required for this project as the propeller direction will need to change in order to provide downward thrust. There is also the potential for waterproof ESCs to be used in order to provide additional security against shorting due to water intrusion. The determination of the amperage of the ESC will be reliant on the other components included within the drone.

Active Battery Life Gauge

Yes, some transmitters have an active battery life gauge included using telemetry which is sent back to the user. This would be beneficial to the project as it would allow the user to know when to bring the drone back and prevent losing communication with the drone either underwater or in the sky.

Conclusion

There are various chemistries of lithium ion batteries used today with clear benefits and disadvantages of each. NMC and LMO power cells allow for quick discharge in high load applications. LiCO₂ and NMC energy cells on the other hand allow for slower discharge but a higher capacity. LiFePO₄ Batteries have a higher C rate than the other chemistries but lack the energy density of the other chemistries. LMO chemistries have a high cycle life but fall short in most other categories. Determining the intended application of the battery is necessary to selecting the correct battery chemistry.

The main task of the Hybrid aerial water drone is to fly out over a body of water, submerge and complete any task desired, then resurface and return to the initial launch location. The importance of control systems and autonomy is to bridge the gap between the energy systems and communication systems of the drone (Figure 1). The main question when designing the controller is, what level of autonomy is necessary for this drone? The answer will direct the design into 3 categories [3]. Fully manual operation of

the drone, fully autonomous operation, and semi-autonomous which is a mix of both. This memo will cover these three approaches regarding available hardware and software.

Background

A flight controller is a necessary part of any flying vehicle. The flight controller's main benefit is providing stability but it also acts as a center for all processing. The flight controller needs to be able to read various incoming signals. These signals can range from manual inputs from a user to onboard sensors relaying current vehicle attitude (Figure 2). The flight controller is no different than a motherboard and processor of a laptop. Everything it does can be simplified into 3 steps. First the FC receives data from its sensors regarding altitude, orientation, and velocity or a desired set point from the user. Second it feeds this data through an algorithm-most commonly a PID [2]. Third, the FC sends out a response to the electric speed controllers that sends power to the actuators. These three steps have different approaches depending on end use. But overall, it defines the characteristics of the drone's flight.

Findings

Hardware: There are plenty of pre-assembled flight controllers available for retail (See *Ref.* [5]). As mentioned a FC is a small computer with attachments depending on its end use. Choosing the right hardware depends on the desired communication protocol, number of channels, physical connection type, voltage range, physical size, required processing power, and cost. Most flight controllers will have built in features such as gyros, barometers, black box memory, and an OSD. It's worth mentioning that the sensors and receivers have communication protocols, and signal types that need to be matched. For a submersible drone the hardware does not differ much between one that flies. The difference can be found in the software parameters adding a buoyancy factor.

Software: The available software isn't restricted to the chosen controller. Most controllers are compatible with more than one software. The content of the software is dependent on what is necessary for the project as mentioned. Fully manual control is not appealing to most consumers. Generally, manual flight control is semi-autonomous because most users want the drone to be stable. Semi-autonomous flight includes features like altitude lock, throttle and angle lock, and return to launch. Fully autonomous control is available however, it is not widely open source which makes it more restrictive on future design requirements. However, signals have a difficult time transmitting through water. Depending on the shareholder's mission the drone potentially needs a fully autonomous algorithm when submerged. Fully autonomous control can range from a simple pre-programmed task, for example, sink to a certain depth, take a reading then return, to active path finding using onboard sensors.

Conclusion

In conclusion there are plenty of options available for controllers. Based on the project mission it would appear that there is a need for a combination of a normal flight control routine and a submersible control routine. Before any requirements any flight routine will do the trick, however, the integration of the submersible side of the mission fully depends on the shareholders requirements. This is where a majority of questions would be asked. Important questions include; does the user want to pilot the drone fully? Does the mission require intermittent manual control? Does the mission routinely occur in the same location? Having these questions answered will direct the priority of the controller's functionality.

Appendix

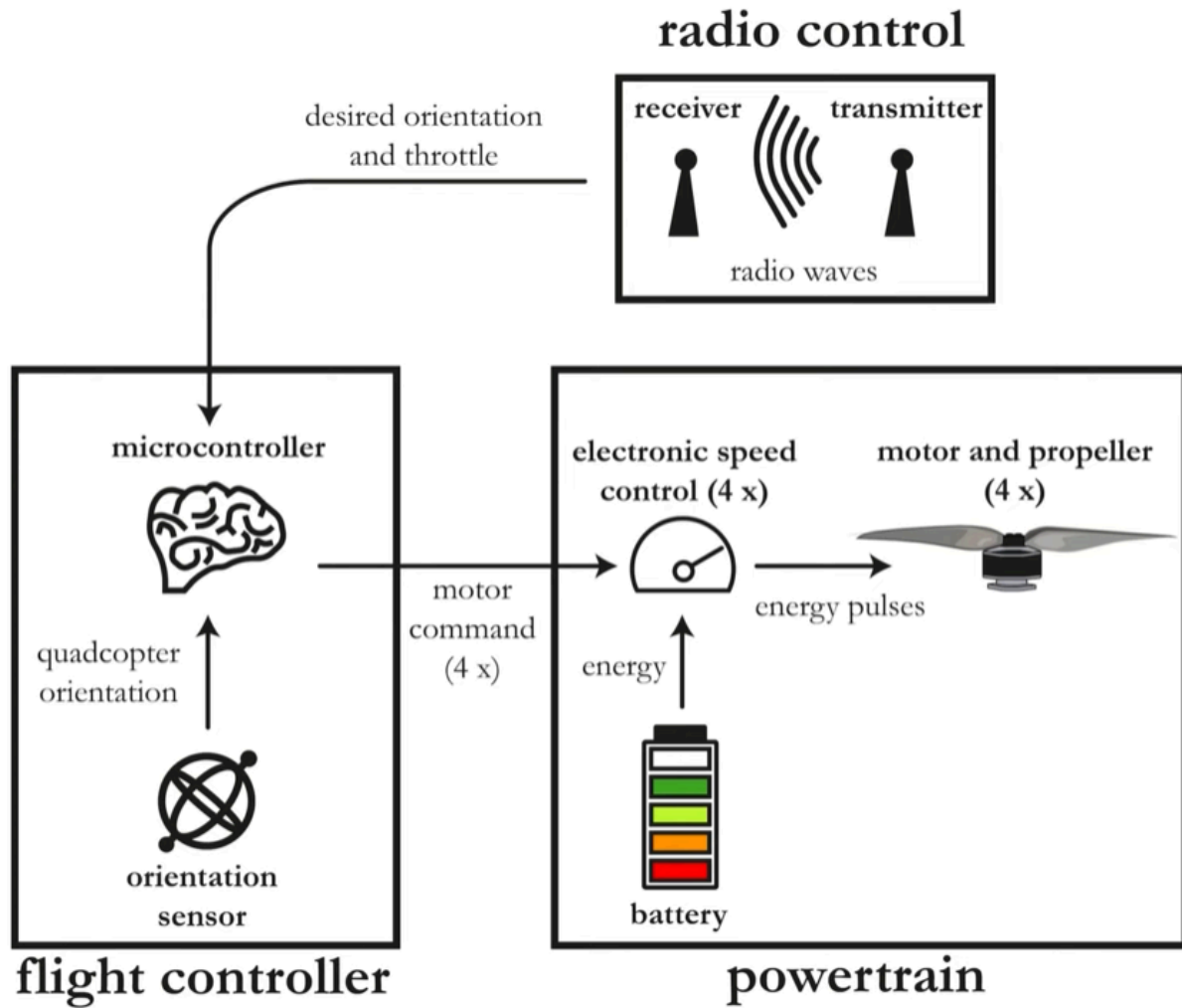


Figure 1: depiction of a flight controller's role in a drone's operation [4].

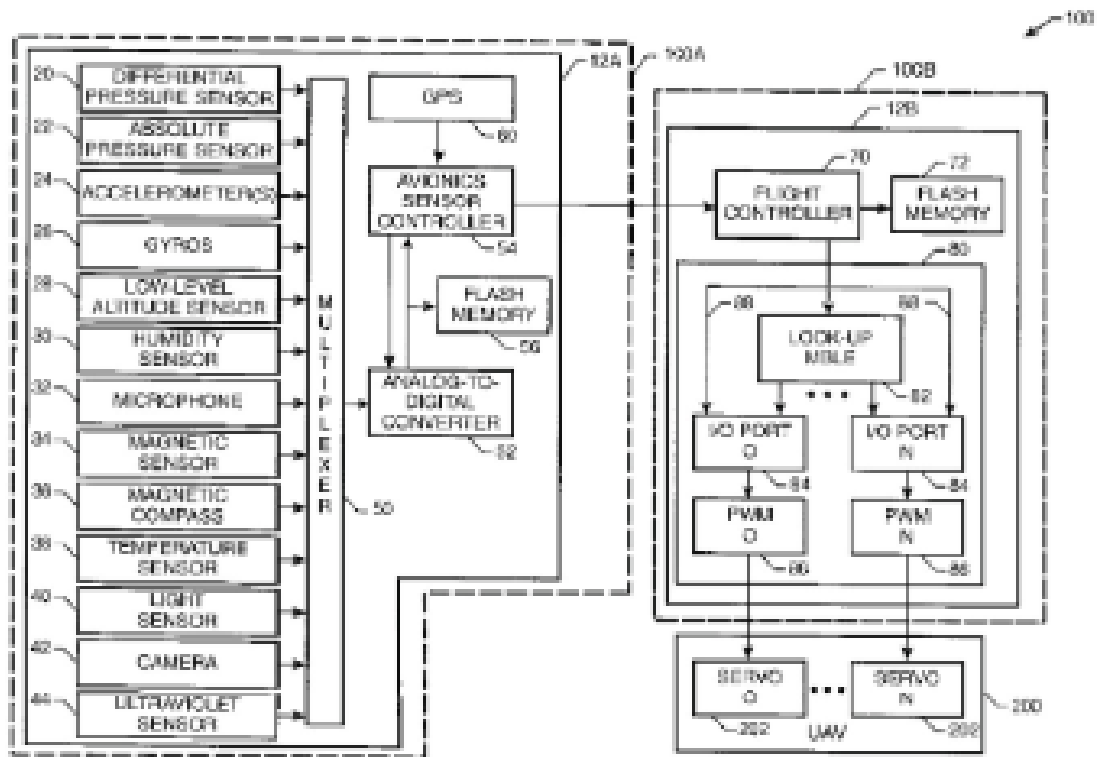


Figure 2: Block diagram example from NASA's Low Weight Flight Controller Design displaying the path from sensors to servos [1].

1. Summary

This technology memorandum evaluates the structural, material, and marine engineering design for a hybrid aerial-underwater drone. The primary goal of this vehicular hull is the conflicting demand of a lightweight frame optimized for aerial flight and a waterproof hull optimized for underwater submersion. This document compares frameworks for material analysis, manufacturing methods, waterproofing, and thermal management. The conclusion section translates these findings into a cultivation of potential parameters for the upcoming requirements phase.

2. Material Analysis

The hull material must maximize structural efficiency across two entirely different load dynamics: aerodynamic flight (minimizing mass) and underwater immersion (Density for submersion).

2.1 Material Property Analysis

To optimize the strength-to-weight ratio while considering thermal conductivity for internal electronics, several material classes are evaluated below.

Material	Manufacturing Process	Density (g/cm ³)	Stiffness (E, GPa)	Tensile Yield Strength (MPa)	Thermal Conductivity (W/m · K)	Waterproof Integrity
Aluminum 6061-T6	CNC Machined	2.7	69	240-276	167	100%
Delrin (Acetal/POM)	CNC Machined	1.41	2.8	55-75	0.28	100%
Acrylic	CNC Machined	1.19	3.2	65-83	0.2	100%
Polycarbonate (PC)	FDM 3D Printed	1.2	2.4 (X-Y) 1.5(Z)	60-70	0.2	Hydrostatically porous (needs post processing)
PETG	FDM 3D Printed	1.27	2.1(X-Y) 1.5(Z)	40-50	0.2	Hydrostatically porous (needs post processing)
ASA	FDM 3D Printed	1.07	2.1(X-Y) 1.4(Z)	45-47	0.17	Hydrostatically porous (needs post processing)
Tough Resin	SLA	1.15	2.2	45	0.17	99.5% (long term risks)

Rigid Resin	SLA	1.5	11	None (Fails at 88 MPa Ultimate Tensile Strength)	0.23	99.9% (better long term compared to tough resin)
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2.2 Manufacturing Analysis: 3D Printed vs Machined

The choice between 3D printing a component versus machining alters its structural integrity, waterproof characteristics, and thermal performance. [2]

3D Printing - FDM/SLA

- **Isotropy:** 3D printed parts are inherently anisotropic. The bond between deposited layers (Z-axis) is weaker than the strength within the print plane (X-Y axis) which is noted in the stiffness column in the table above. This means that 3D printed materials have the potential to be completely waterproof depending on the printing settings [4] . However, some are less porous.
- **Micro-Porosity:** Fused Deposition Modeling (FDM) leaves microscopic voids between layer lines. Water can leak through these boundaries over time, which can lead to eventual internal flooding. Selective Laser Sintering (SLS) also produces porous structures. Stereolithography (SLA/Resin) provides the best fluid density and waterproofing but can be very brittle. [3] .
- **Surface Roughness (Ra):** Printed surface finishes are rough. This could wear out the elastomeric O-rings or prevent them from seating correctly, as fluid can get through the seal through the microscopic voids of the print lines. Post-processing (vapor polishing, epoxy coating) would be highly encouraged for sealing surfaces.
- **Prototyping Flexibility:** 3D printing allows for complex geometries, internal hollow fluid channels, and rapid design iterations at low cost.

CNC Machining (Manufacturing)

- **Mechanical Isotropy:** Machined components are *isotropic*, meaning the material has uniform properties in all directions. This means that the entire surface/build will react predictably to all stimuli and factors
- **Fluid Density:** Solid billet stock (Aluminum 6061-T6 or Delrin) contains zero internal voids, ensuring a 100% reliable fluid barrier across the material walls.

- **Surface Finish:** Precision machining easily achieves the low surface roughness tolerances ($R_a \leq 0.8 \mu\text{m}$ for static joints and up to $16 \mu\text{m}$ for dynamic joints) required for solid O-ring gland compression and any shaft interfaces.
- **Weight and Cost Trade Offs:** Creating thin-walled, lightweight enclosures require long machine times, high material waste, and higher production costs. Machined materials also tend to be heavier.

2.3 Thermal Energy Dissipation

Naturally, sealed hulls will result in thermal insulation, which can put all electronics inside at risk. The structural skin must dissipate heat produced by these electronic components. The heat transfer rate through the hull boundary is governed by Fourier's Law:

$$\dot{Q} = \frac{k \cdot A \cdot \Delta T}{t}$$

Where $\dot{Q}$ is heat transfer rate, k is material thermal conductivity, A is surface area, ΔT is temperature difference, and t is thickness. Machined metals (Aluminum) provide a direct, high-conductivity thermal path ($k = 167 \text{ W/m} \cdot \text{K}$), whereas printed polymers ($k \approx 0.2 \text{ W/m} \cdot \text{K}$) trap heat, requiring metallic heat-sink penetrations.

3. Waterproofing and Sealing Technologies

Operating in marine environments requires waterproofing joints, shafts, and managing saltwater corrosion [1].

3.1 Interface Sealing

- **Static O-Rings:** Elastomeric O-rings (Buna-N for fuel/oil resistance or EPDM for UV and saltwater stability are some examples) placed in between joints and connections offer reliable fluid barriers. There are different strength level O-Rings for varying pressure loads [1].
- **Rotary Seals vs. Potting:** To seal potential spinning propeller shafts, or anywhere that O-rings or seals could be utilized, there are two potential solutions.
 - *Lip Seals / Rotary O-rings:* Provide waterproof containment, but can introduce rotational friction, which can result in reduction of battery life during the aerial flight phase.
 - *Potting:* Potting is the process of adding a liquid (epoxy) capable of solidifying around electrical components to protect them. This eliminates shaft friction but introduces other negatives such as the inability to make changes to electrical components and could also result in a heavier hull design.
- **Waterproof and Corrosion Surface Coatings:** To mitigate micro-porosity of 3D-printed hulls, external surfaces must be post-treated with low-viscosity waterproof coatings such as epoxy (more expensive) which does well against water and can fit tight

spaces, but can be rigid, or silicone, which is strong and flexible. Coatings could also be used to prevent salt water corrosion.

4. Geometry & Fluid Flow Analysis

The geometry must be optimized for external fluid flow, minimizing drag in both air and water, and mitigating water-entry impacts.

4.1 Fluid Density Scaling & Drag Considerations

The drag force acting on the vehicle profile is governed by:

$$F_d = \frac{1}{2} \rho v^2 C_d A$$

Where ρ is fluid density, v is velocity, C_d is the drag coefficient, and A is the frontal area [5].

4.2 Geometric Profiles

- **Tear-Drop / Airfoil Shape:** Minimizes drag coefficient during forward translation in both air and water but can be a problem for moving in any other direction. Also, if the drone transitions vertically into the water, a horizontal tear-drop shape has a large cross-sectional area, resulting in potentially harmful impact loads on the hull when hitting the water's surface. [5]
- **Cylindrical Shape:** Has better capabilities for reducing C_d in forward and backwards directions, however, it still experiences drag in omnidirectional movements. It also allows a better space for internal electronics [5].
- **Spherical Shape:** Most balanced shape as it allows for uniform C_d in omnidirectional movement. Has slightly higher C_d values but is the most balanced in terms of performance.

5. Key Takeaways

- **Waterproofing and Materials:** Prioritizing isotropic materials for dry regions. If utilizing 3D printing, especially FDM, post processing should be done for waterproofing the printed structure. Polycarbonate or Acrylic should be considered for the main central hull to achieve a watertight enclosure.
- **Stiffness vs Strength:** Factors should be considered for how hard the drone will impact water when choosing a material. Stiffness should take priority if the drone gets eased into the water surface, whereas tensile yield strength would be prioritized for rough impacts. Enclosure walls should remain rigid, however, to optimize pressure for the proper function of O rings in case of their utilization.

- **Surface Roughness:** Any point of contact of the hull surface where the elastomer seal (O-ring) touches should meet a specific tolerance to prevent liquid weeping through. Post processing procedures should be considered.
- **Thermal Management:** Polymers act as insulators whereas some CNC machined materials like aluminum have high conductivity. Design should utilize high conductivity sections or pathways that can vent the heat produced inside the hull from electronics directly to the water on the outside.
- **Propulsion Sealing:** The team should decide on proper sealing for rotors depending on design. This can include rotary lip seals where friction could be a risk, or epoxy potted stators that can introduce drag in water, and potentially trap components permanently.
- **Geometry:** To mitigate the significant differences of density induced drag (water vs air), the hull should use appropriate shape depending on how the drone is designed to transverse the air and water (forward motion or omnidirectional).

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